

Apostol Chapter 1 - Exercise 1

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If $(a, b) = 1$, then (a^n, b^k) for all $n \geq 1, k \geq 1$.

Suppose that $(a, b) = 1$.

By the Fundamental Theorem of Arithmetic, we can write

$$a = \prod_{i=1}^{\infty} p_i^{\alpha_i} \quad \text{e} \quad b = \prod_{i=1}^{\infty} p_i^{\beta_i}$$

Then, we know that,

$$1 = (a, b) = \prod_{i=1}^{\infty} p_i^{c_i},$$

where $c_i = \min(\alpha_i, \beta_i)$. Since the left side is 1, and the right side is a product of primes, we necessarily have that each term in the product must be equal to 1. That is, $c_i = 0$, for every $i \geq 1$. But $c_i = 0$ means that at least one $\alpha_i = 0$ or $\beta_i = 0$. Thus, for each $i \geq 1$, $\alpha_i = 0$ or $\beta_i = 0$.

Let $n \geq 1$ and $k \geq 1$. We can also write,

$$a^n = \prod_{i=1}^{\infty} p_i^{n\alpha_i} \quad \text{e} \quad b^k = \prod_{i=1}^{\infty} p_i^{k\beta_i}$$

Then,

$$(a^n, b^k) = \prod_{i=1}^{\infty} p_i^{d_i},$$

where $d_i = \min(n\alpha_i, k\beta_i)$. But we know that for each $i \geq 1$, either $\alpha_i = 0$ or $\beta_i = 0$. Thus, for each $i \geq 1$, either $n\alpha_i = 0$ or $k\beta_i = 0$. That is, for every $i \geq 1$, $d_i = 0$. Therefore, $(a^n, b^k) = 1$.